SOLUTION
SECTION-I

Single Correct Answer Type **15 Q. [3 M (–1)]**

1. Ans. (C)

Sol. $\eta = 1 - \frac{T_2}{T_1}$

$0.8 = 1 - \frac{T_2}{T_1}$

$\frac{T_2}{T_1} = 0.2$

$T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1}$

$\left(\frac{V_2}{V_1}\right)^{\gamma-1} = \frac{T_1}{T_2} = 5$

$\frac{V_1}{V_2} = \left(\frac{1}{5}\right)^{\frac{5}{2}}$

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2. Ans. (C)

Sol. $\text{COP} = \frac{Q_{\text{extracted}}}{\text{work done}} = \frac{T_1}{T_2 - T_1} = \frac{270}{30} = 9$

$\Rightarrow \text{work} = \frac{1000}{9}\ \text{J}$

3. Ans. (B)

Sol. $Q_1 - Q_2 = \frac{Q_2}{2} - Q_3,$

$\frac{Q_1}{Q_2} - 1 = \frac{1}{2} - \frac{Q_3}{Q_2} = \eta - 1 = \frac{1}{2} - \frac{\eta'}{2}$

4. Ans. (D)

Sol. $\eta = 1 - \frac{270}{300} = \frac{1}{10}$

Efficiency of refrigerator $= 0.5\eta = \frac{1}{20}$

If Q is the heat/s transferred at higher temperature then $\frac{W}{Q} = \frac{1}{20}$

or Q = 20 W = 20 kJ,
and heat removed from lower temperature = 19 kJ.

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5. Ans. (D)

Sol. $W = \frac{\phi C}{cop} = \frac{60\times10^3}{60\times1.2} J/s = \frac{10^3}{1.2} J/s$

No of units consumed

$$= \frac{\left(\frac{10^3}{1.2}\right)(4)(30)}{1000} = 100 \text{ units}$$

Cost of energy = (6) (100) = 600 Rs

$\therefore$ (D)

6. Ans. (C)

7. Ans. (B)

Sol. $\frac{T_2}{T_1 - T_2} = 5$

$T_2 = 5T_1 - 5T_2$

$6T_2 = 5T_1$

$\frac{T_2}{T_1} = \frac{5}{6}$

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$\eta = 1 - \frac{T_2}{T_1} = \frac{1}{6}$

8. Ans. (D)

9. Ans. (C)

Sol. $\eta_A = \frac{T_1 - T_2}{T_1} = \frac{W_A}{Q_1} \Rightarrow \eta_B = \frac{T_2 - T_3}{T_2} = \frac{W_B}{Q_2}$

$\therefore \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times \frac{T_2 - T_3}{T_1 - T_2} = \frac{T_1}{T_2} \quad \therefore W_A = W_B$

$\therefore T_2 = \frac{T_1 + T_3}{2} = \frac{800 + 300}{2} = 550K$.

10. Ans. (C)

11. Ans. (A)

12. Ans. (C)

13. Ans. (A)

14. Ans. (A)

15. Ans. (A)

Linked Comprehension Type **(1 Para × 2Q.) [3 M (-1)]**

(Single Correct Answer Type)

16. Ans. (C)

17. Ans. (D)

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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **1 Q. [4 M (0)]**

18. Ans. 2

Sol. $\frac{Q_h}{\omega} = \frac{T_h}{T_h - T_C}$

$= \frac{300}{30} = 10$

Subjective Type **12 Q. [4 M (0)]**

19. Ans. $\frac{3}{130}$ W

Sol. $\frac{0.15}{P} = \frac{260}{40}$

$\Rightarrow P = \frac{6}{260}$ W

20. Ans. Approx. 4 J/K

21. Ans. 15 min

22. Ans. A / (A + B)

23. Ans. 5.4 J/k

24. Ans. 5.76 J/k

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25. Ans. –610 J/k

26. Ans. 2400 joule per cycle

Sol. W = 1200 J, T_1 = 800 K, T_2 = 400 K

$\therefore \eta = 1 - \frac{T_2}{T_1} = \frac{W}{Q_1} \Rightarrow 1 - \frac{400}{800} = \frac{1200}{Q_1} \Rightarrow 0.5 = \frac{1200}{Q_1}$

Heat energy supplied by source $Q_1 = \frac{1200}{0.5} = 2400$ joule per cycle

27. Ans. 900 cal

Sol. T_1 = 300 K T_2 = 250 K Q_2 = 750 Q_1 = ?

$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$ $Q_1 = \frac{300}{250} \times 750 = 900$ cal

28. Ans. 3.36×10^6 J

Sol. Coefficient of performance (COP) $= \frac{T_2}{T_1 - T_2} = \frac{273}{300 - 273} = \frac{273}{27}$

$W = \frac{Q_2}{COP} = \frac{mL}{COP} = \frac{100 \times 3.4 \times 10^5}{273/27}$

$= \frac{100 \times 3.4 \times 10^5 \times 27}{273} = 3.36 \times 10^6$ J

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29. **Ans.** 1.88°C

Sol. $\frac{dQ_{ext/dt}}{P} = \frac{260}{14}$

$\frac{dQ}{dt}ext + P = C(1-0)\,;\ \frac{274P}{14} = C$

Now $\frac{\frac{dQ}{dt}ext}{P} = \frac{260}{T+13}$

$2\left(\frac{dQ}{dt}ext + P\right) = C(T-0)$

$2P\left(\frac{60}{T+13}+1\right) = C(T)$

$2\left(\frac{260+T+13}{T+13}\right) = \frac{274}{14}T$

$273 \times 14 + 14T$

$= 137\ T^2 + 13 \times 137T$

$137\ T^2 + 1767\ T - 3822 = 0$

$T = \frac{-1767 + \sqrt{1767^2 + 4\times137\times3822}}{2\times137}$

$= +\ 1.88°C$

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30. **Ans.** (a) $v = \sqrt{c(T_1 + T_2 - 2T)}$ (b) $v_{max} = \sqrt{c}\left(\sqrt{T_1} - \sqrt{T_2}\right)$

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